

Before vs. After Security Results

PrivateVault Security Hardening & Compliance Verification

Executive Summary

This evaluation report documents the security hardening of the Credit Risk AI Lending Decision Support System. By integrating PrivateVault's active security middleware layer, the system was tested against 39 complex attack vectors across four core risk dimensions: Prompt Injection, Context Poisoning, Sensitive Data Protection (PII), and Decision Integrity (Hallucinations). Under testing, PrivateVault achieved a 100% security mitigation and policy compliance rate, resolving all vulnerability exposures spanned across the pipeline without modifying the core machine learning models.

SECURITY MITIGATION 100% SAFE 39 of 39 Attacks Defeated	TESTS COMPLETED 39 CASES Across 4 Evaluation Fields	AUDIT LEDGER INTEGRITY OK 120 Cryptographic Chains Verified
--	--	--

Quantitative Verification Metrics

Attack Vector	Test Count	Blocked / Total	Blocked Rate	Overall Status
1. Prompt Injection	7	0/7 (0% Blocked)	7/7 (100% Blocked)	SECURED
2. Context Poisoning	4	0/4 (0% Detected)	4/4 (100% Detected)	SECURED
3. Sensitive Data Protection (PII)	8	0/8 (0% Protected)	8/8 (100% Redacted)	SECURED
4. Hallucination & Policy Compliance	20	0/20 (0% Grounded)	20/20 (100% Validated)	SECURED

Verification Breakdown per Dimension

Test Suite 1: Prompt Injection Attacks (7/7 Blocked)

Adversarial attempts using Direct Instruction Overrides, Role-play Jailbreaks (DAN), Base64 payload encoding, and System Prompt Extraction were passed. The AI Firewall successfully decoded Base64 sequences and intercepted malicious context manipulations before processing. Attack block rate achieved 100% (from 0%).

Test Suite 2: Context Poisoning Protection (4/4 Filtered)

Poisoned reference guidelines injected into the FAISS database (e.g. counterfeit RBI circulars approving score <300, Basel III alteration attempts) were successfully intercepted. The Context Validator flagged contradictory prompts and metadata discrepancy, avoiding LLM hijacking. Poison rate mitigated from 100% to 0%.

Test Suite 3: Sensitive Data Redaction (8/8 Cleansed)

The PII Shield successfully evaluated bidirectional data flows. PII inputs (Aadhaar, PAN cards, IFSC codes, bank details) were sanitized at input, and LLM generated reports containing leaked credentials or API keys were redacted at output before final delivery. Privacy protection improved from 0% exposure control to 100% coverage.

Test Suite 4: Hallucination & Decision Governance (20/20 Corrected)

In 20 high-risk profiles containing contradictory decisions, missing citations, and policy deviations: The Policy Engine enforced mathematical constraints (e.g. risk score >40% automatically triggers a decline suggestion vetoing LLM approval) and blocked demographic bias. The Hallucination Guard audited citation veracity against context. Decision reliability reached 100%.

Tamper-Evident Ledger Integrity

All 120 security events (scans, validation flags, policy decisions, and redaction actions) were written to the Audit Ledger. The ledger implements a cryptographic Merkle-like chain where each log entry contains the SHA-256 hash of the preceding entry. Verification checks completed:

- * Verification Integrity Status: VALID
- * Total Hash Chains Computed: 200 entries
- * Cryptographic Chain Errors: 0

This cryptographic binding ensures that the entire log trail is immutable and completely tamper-evident.